



EAST WEST UNIVERSITY

CSE302: Database Systems

Section: 12

Summer-2026

Project Report

Project Title: University Portal Management System.

Project URL:

<https://github.com/malihaesha28-lan/DatabaseProject1> [https://github.com/Salman-Saad-Mahmud/Database Project 01](https://github.com/Salman-Saad-Mahmud/Database_Project_01) [https://github.com/Zahedul-zane/EWU Portal management system/tree/main](https://github.com/Zahedul-zane/EWU_Portal_management_system/tree/main)

Submitted by:

Student ID	Student Name	Contribution Percentage
2023-2-60-079	Maliha Mumtaj Esha	32%
2023-3-60-621	Salman Saad Mahmud	34%
2024-3-60-561	Zahedul Islam	34%

Submitted to:

Antu Chowdhury

Lecturer

Department of Computer Science & Engineering

1. Project Description

1.1 Overview

The University Portal Management System is a database application designed to form a core for automating academic, administrative, and financial functions of a University. The system stores data about students, faculty, section, departments, course, payment and enrollment thus forming an integrated system for course registration, academic advising, assessments, and administrative procedures.

1.2 Real-World System Simulated

The project simulates the university level academic and administrative information system used in tertiary educational institutions. The system is utilized for various academic and administrative purposes including:

. Academic Advising and Course Registration–

This includes pre-advising (student requests for course registration per semester) and enrollment of students into specific sections of a course.

.Curriculum and Prerequisites–

This refers to the management of curriculums and their prerequisites which are followed in the course registration process.

.Faculty and Department Management–

This entails management of the departments, their location (office addresses), heads of departments, and faculty – including their course assignments and time-tables.

.Student Assessment–

Maintaining information on student's assessment results including students midterm marks, final marks, grades, and CGPA.

.Managing Tuition and Fees Payments–

This includes recording of payments and their clearance, with the following information stored per semester: payment type, transaction ID, payment date, and status.

.Notices and Communications–

Posting of notices for members of staff and students by the faculty.

1.3 Main Purpose and Data Handled

Primary Purpose “The University Portal Management System will allow a university to computerize administrative and academic processes.” The main purpose of this project is to provide automation and centralization of the majority of the essential processes. It will be crucial in ensuring the elimination of unnecessary paperwork by substituting it with technological processes. In addition, this system will guarantee such advantages as elimination of inconsistent data and simplification of academic and administrative rules, increased efficiency in the day-to-day operations, and improved accessibility of the information to students, faculty, and administration.

2.Key Features

The **University Portal Management System** provides a centralized platform that supports various academic and administrative functions tailored to students, faculty, and administrators. The features have been grouped according to activities that the entities perform on the portal as discussed below:

Student Features

The student activities include submitting pre-advising requests with preferred course selections and timestamps enrolling in classes according to their approval status and room capacities, viewing progress reports consisting of credit hours, assignment grades (mid and final terms), and letter grades, processing payments and viewing their statuses (Payment_Status), and accessing departmental announcements published on the portal.

Faculty Features

The faculty activities include viewing their class schedules, classroom allocations (Room_No), and advising students on their pre-advised requests.

System & Administrative Capabilities

The faculty members can process and approve student requests and pre-advising applications, manage student grades and publish announcements with contents. Meanwhile, the head faculty activities involve managing course offerings at the department level. Other activities include enforcing prerequisite relationships (Requires relationship) among students and courses in terms of enrollment restrictions, enforcing capacity restrictions on sections considering the room capacities, recording all payments in the portal and managing financial accounts with timestamps and establishing relationships (works_in) between the departments, faculty members and students enrolling in the institution's courses.

3. Database Design

3.1 E-R Model

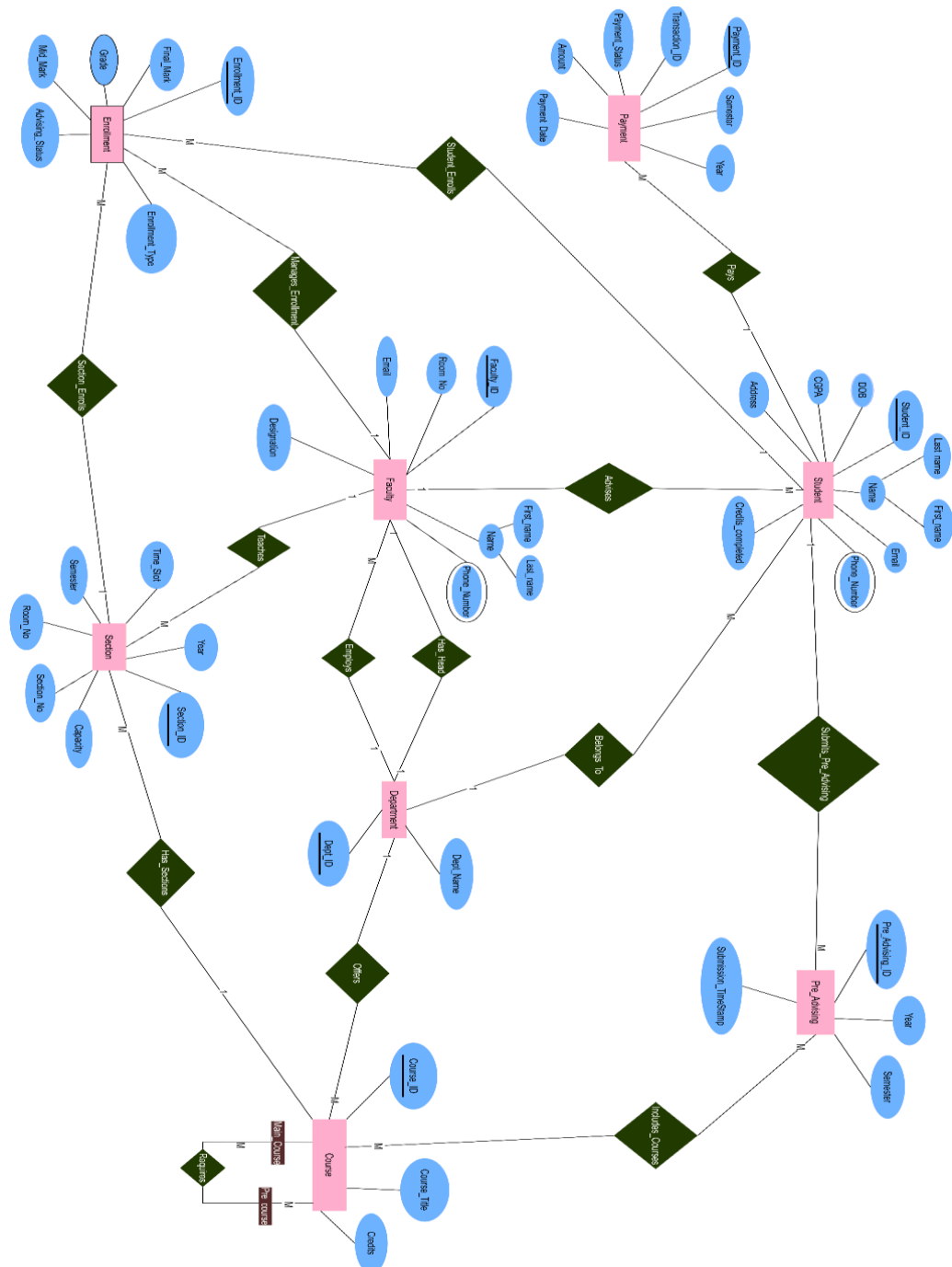


Figure 1: ER diagram on University portal management system .

3.2 Relational Data Model

Student(Student_ID, First_name, Last_name, E-mail, Address, DOB, Faculty_ID(FK), Dept_ID(FK))

Student_PhoneNum(Student_ID(FK), Phone_Number1, Phone_Number2)

Faculty(Faculty_ID, First_name, Last_name, Designation, Room_No, E-mail, Dept_ID(FK))

Faculty_PhoneNum(Faculty_ID(FK), Phone_Number1, Phone_Number2)

Department(Dept_ID, Dept_Name, Head_Faculty_ID(FK))

Course(Course_ID, Course_Title, Credits, Dept_ID(FK))

Course_Prerequisite(Course_ID(FK), Pre_Course_ID(FK))

Enrollment(Enrollment_ID, Enrollment_Type, Advising_Status, Mid_Mark, Final_Mark, Grade, Section_Id(FK), ManagedBy_Faculty_Id(FK), Student_ID(FK))

Pre-advising(Pre_Advising_ID, Submission_TimeStamp, Semester, Year, Student_ID(FK))

IncludedCourse(Pre_Advising_ID(FK), Course_ID(FK))

Section(Section_Id, Section_No, Time_Slot, Room_No, Capacity, Semester, Year, Course_ID(FK), Faculty_ID(FK))

Payment(Payment_Id, Transaction_Id, Payment_Status, Amount, Semester, Year, Payment_Date, Student_ID(FK))

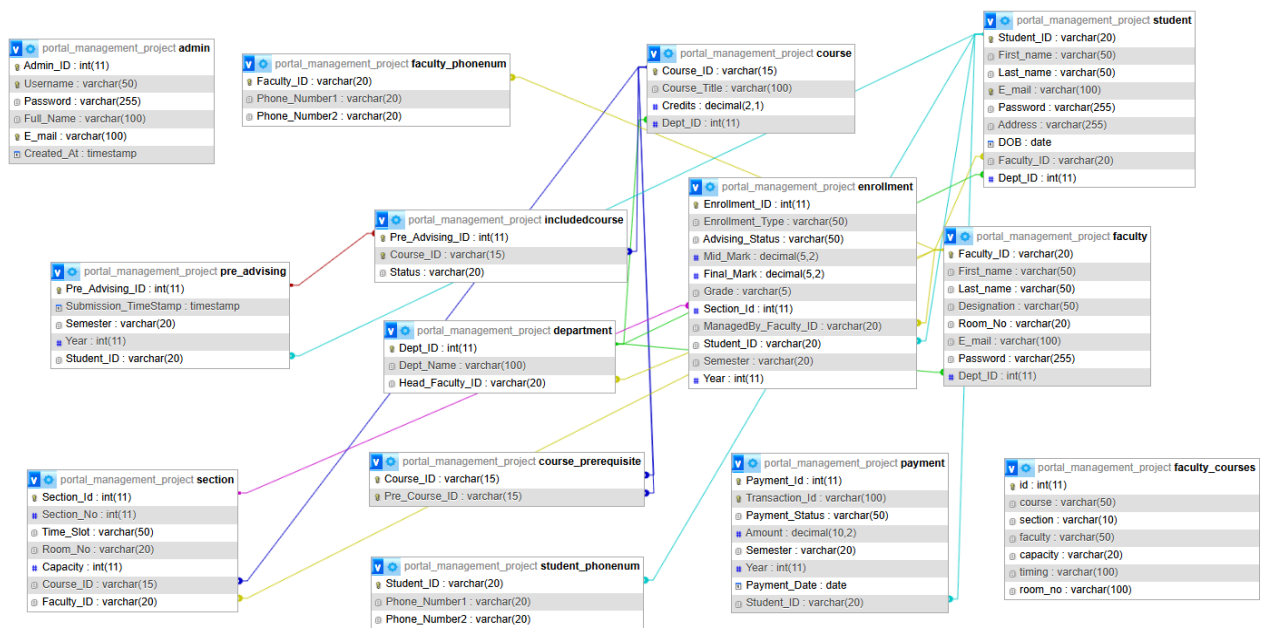


Figure : Schema Diagram.

4. Tools & Technologies Used

To conduct this kind of database related problem first we used XAMPP as SQL query software. To accomplish this fully into a GUI based environment we used HTML as front-end code and as back-end code we used PHP. We related the PHP codes with the database sql command to conduct the system perfectly.

Front End Technologies HTML5: This technology serves as the foundation for building the user interface in the portals for student, faculty and administrator roles with form elements, modals and role switching components.

CSS3: Style sheet for the purpose of styling, layout design using grid/flex box, responsive view ports, alert component styling and dynamic tab navigation.

JavaScript: It handles the front-end interactions, DOM manipulations, dynamic tab navigation based on user roles and automatic single click demo credential auto-filling (main.js).

Vector Graphics : It is used in the controls of the interface to render the scalable graphics performing actions like sign-in.

Back End Technologies PHP: This technology serves as the back-end scripting language to handle HTTP request processing, user authentication, application views routing and role-based access control.

PHP Data Objects (PDO): This technology is used for abstraction for running the parameterized queries and prepared statements to avoid SQL injections.

PHP Session Management: This technology is used to manage the login state of the users (session_start), maintain the context of users in the session and validate the role authorization (student, faculty, admin).

Bcrypt Hashing Algorithm (password_verify): Password hashing and password verification of user accounts.

Database and Infrastructures MariaDB Server: Relational Database Management System (RDBMS) configured with the character set utf8mb4_general_ci for storing relational entities such as students, faculty members, courses, enrollments and payments.

phpMyAdmin: Web-based database management system to run the SQL scripts, configure foreign keys, create primary key index and export database schema.

XAMPP: Local web server environment to host PHP application engine and manage HTTP client-server communications.

5. Role Assignment

1. System Administrator(Admin): Full operational and administrative access across the entire EWU portal database. Also can access financial portal and billing management.

They can create, update and deactivate(remove) user accounts like Students and Faculty. Manages database backups, overrides enrollment caps, section limits, or system locks during technical issues. Also can approve course offerings, section allocations for upcoming semesters, verifies student bank transactions, online payment gateways and update student Payment_Status flags to grant financial clearance for advising. Finally ensures security protocol compliance on the overall system.

2. Faculty member and department head: Faculty members can manage advising, can enter grade and access the course sections. On the other hand, the department head can do administrative and supervisory access.

Faculty members can access assigned section class rosters and student details. Also can inputs and updates assessment marks (midterm, final marks). Also can serve as an academic advisor to review, guide, and approve student advising/pre-advising requests. Whereas department head reviews and resolves student academic appeals, prerequisite waivers, or course overload requests.

3. Student: Can access personal academic and financial portal access. They can submit course selections during pre-advising and complete formal enrollments. Views section schedules, assigned faculty, class locations, and time slots. Checks academic performance, individual assessment marks, grades, and completed credits. Finally, they can submit tuition payments and track transaction statuses.

6. GUI Screenshot

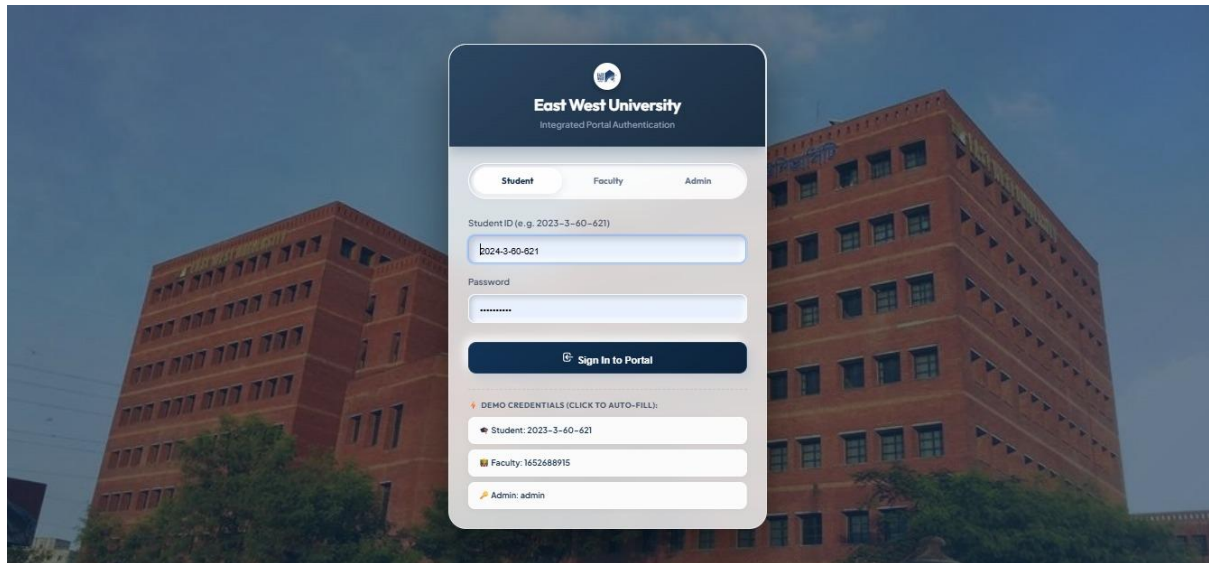


Figure 2 : Login Interface .

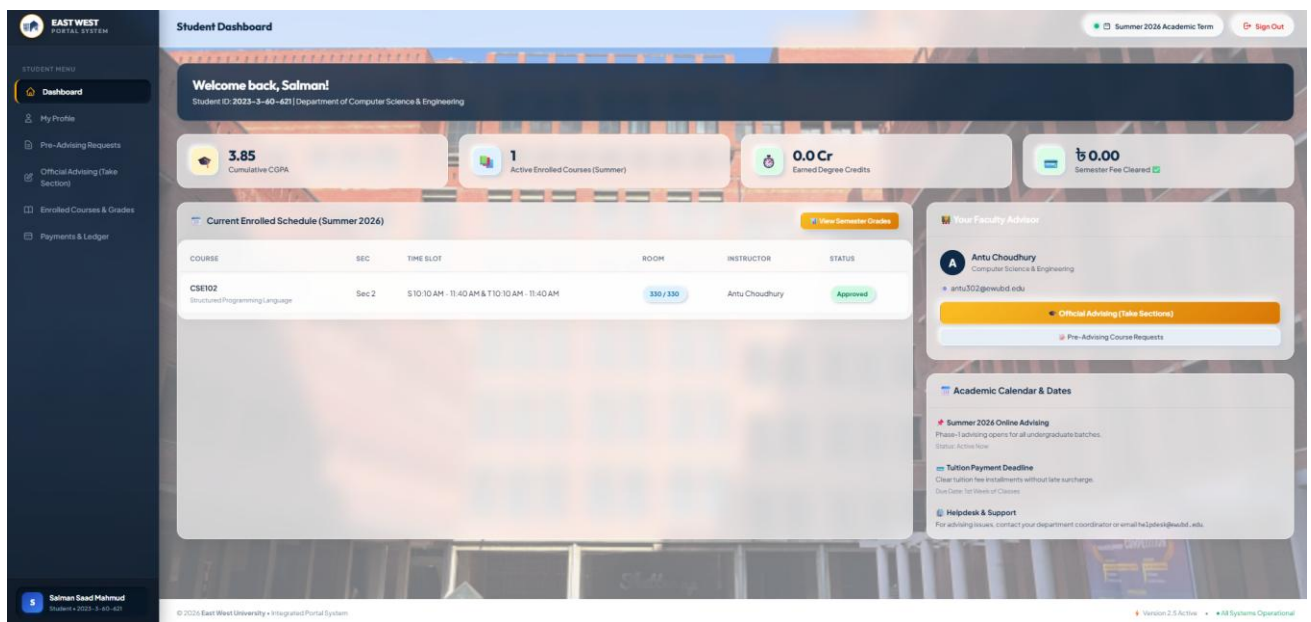


Figure 3: Student Dashboard .

EAST WEST
PORTAL SYSTEM

STUDENT MENU

- Dashboard
- My Profile**
- Pre-Advising Requests
- Official Advising (Take Section)
- Enrolled Courses & Grades
- Payments & Ledger

Salman Islam

Student Profile

Summer 2026 Academic Term Sign Out

Student Profile & Settings

Manage your official record, contact details, and account security.

Academic Identification

S

Salman Islam

2024-3-60-621

Department:
Computer Science & Engineering

Institutional Email:
2024-3-60-621@std.ewu.edu

Date of Birth:
N/A

Faculty Advisor:
Nishat Tasnim

Update Profile Information

Residential Address

Bangladesh

Primary Contact Number (*)

01949304965

Secondary Contact Number (Optional)

2024-3-60-621

Security Settings

Change Password (Leave blank to keep current)

Save Profile Changes

Figure 4: Student Profile.

EAST WEST
PORTAL SYSTEM

STUDENT MENU

- Dashboard
- My Profile
- Pre-Advising Requests**
- Official Advising (Take Section)
- Enrolled Courses & Grades
- Payments & Ledger

Salman Saad Mahmud
Student • 2023-3-60-43

Student Advising & Course Enrollment

Summer 2026 Academic Term Sign Out

Pre-Advising & Course Enrollment

Submit course requests to your advisor, then enroll in sections once they are approved.

Official Advising & Section Selection is Live!

Once your advisor approves your requested courses, open Official Advising to select your section and schedule.

Go to Official Advising

YOUR ASSIGNED ADVISOR

Antu Choudhury

Computer Science & Engineering antu2023@ewu.edu

0 Awaiting Approval

2 Advisor Approved

Step 1 — Request Courses from Advisor

Select the courses you want to take this semester. Your advisor Antu Choudhury will review and approve each course.

COURSE	CR.	PRE-REQ
☐ CSE302 Database System	4.5	None
☐ CSE101 Discrete Mathematics	3.0	None
☐ CSE102 Structured Programming Language	3.0	None
☐ CSE103 Structured Programming Language	4.0	None
☐ CSE104 Discrete Mathematics & Logic	3.0	CSE103

Send to Advisor for Review

Step 2 & 3 — Approval Status & Section Enrollment

Summer 2026 — Form #4 Submitted 19 Aug 2026, 10:07 PM

2 courses

CSE302 Database System 4.5 Cr **Advisor Approved**

Approved — no class sections scheduled yet. Please check back later.

CSE102 Structured Programming Language 3.0 Cr **Advisor Approved**

Enrolled — Section 2

5:10:10 AM - 11:40 AM & T 10:10 AM - 11:40 AM • 330 / 330

Figure 5: Student Advising & Course Enrollment.

EAST WEST
 PORTAL SYSTEM

STUDENT MENU
[Dashboard](#)
[My Profile](#)
[Pre-Advising Requests](#)
Official Advising (Take Section)
[Enrolled Courses & Grades](#)
[Payments & Ledger](#)

Salman Islam
 Student • 2024-3-60-421

Official Course Advising & Section Registration
 Select your class sections and build your conflict-free timetable from courses approved by your advisor.

ADVISOR
Nishat Tasnim

APPROVED CR
7.0 Credits

REGISTERED
7.0 Cr

Search by course code (e.g. CSE101), faculty (e.g. MAR, MSHQ), or room...

CSE101 Discrete Mathematics
3.0 Credits

Approved by Advisor

Registered In Section 4
 T 08:30 AM - 10:00 AM & R 08:30 AM - 10:00 AM • Room: AB2-201 / 372 (SEIP) • Faculty: Dr. Ahmed Hasan
[Change / Drop Section](#)

CSE103 Structured Programming Language
4.0 Credits

Approved by Advisor

Registered In Section 2
 M 10:10 AM - 11:40 AM & W 10:10 AM - 11:40 AM • Room: 325 / 372 (SEIP Lab) • Faculty: Dr. Ahmed Hasan
[Change / Drop Section](#)

Registered Weekly Routine (Summer 2026)

CSE101 • Sec 4
3.0 Cr
 T 08:30 AM - 10:00 AM & R 08:30 AM - 10:00 AM
 AB2-201 / 372 (SEIP)

CSE103 • Sec 2
4.0 Cr
 M 10:10 AM - 11:40 AM & W 10:10 AM - 11:40 AM
 325 / 372 (SEIP Lab)

Total Registered Credits: **7.0 Cr**

Figure 6: Official Course Advising & Section Registration.

EAST WEST
 PORTAL SYSTEM

STUDENT MENU
[Dashboard](#)
[My Profile](#)
[Pre-Advising Requests](#)
[Official Advising \(Take Section\)](#)
Enrolled Courses & Grades
[Payments & Ledger](#)

Salman Islam
 Student • 2024-3-60-421

Academic Marksheet & Semester Grades
 View your term grade reports, SGPA breakdown, cumulative progress (CGPA), and official marks by semester.

Academic Advisor: Prof. Nishat Tasnim (Computer Science & Engineering)

CUMULATIVE CGPA
3.85 / 4.00
 Dean's HonorList

EARNED CREDITS
0.0 / 140 Cr
 0% Degree Completed

SEMESTERS COMPLETED
0
 Total 1 Semesters Enrolled

REGISTERED COURSES
2
 Across all terms

All Semesters Summer 2026 Current

Summer 2026 Marksheet
 Registered Courses: 2 | Total Credits: 7.0 Cr

Semester SGPA: N/A Earned: 0.0 Cr Term In Progress Active Semester

#	COURSE CODE & TITLE	CREDITS	SECTION & SLOT	INSTRUCTOR	MID MARK (30)	FINAL MARK (70)	LETTER GRADE	GRADE POINT	STATUS
1	CSE101 Discrete Mathematics	3.0	Sec 4 T 08:30 AM - 10:00 AM & R 08:30 AM - 10:00 AM Room: AB2-201 / 372 (SEIP)	Dr. Ahmed Hasan ahmed.hasan@ewubd.edu	—	—	N/A	—	Approved

Figure 7 : Academic Marksheet & Semester Grades.

STUDENT MENU

- Dashboard
- My Profile
- Pre-Advising Requests
- Official Advising (Take Section)
- Enrolled Courses & Grades
- Payments & Ledger**

Selman Saad Mahmud

Student • 2023 - 2-40-423

Semester Fee Assessment & Payment

Semester Fee Assessment & Payments

Course-based tuition calculation, online semester fee payment submission, and financial clearance tracking.

18,000.00

Summer 2026 Fee (3 Credits)

18,000.00

Verified Paid (Summer Term)

0.00

Pending Admin Verification

0.00

Outstanding Due Balance

Summer 2026 Course Fee Assessment

Paid (Cleared)

COURSE DETAILS	SECTION	CREDITS	RATE / CR	COURSE TUITION
CSE102 Structured Programming Language	Sec 2	3.0 Cr	5,500.00	16,500.00
Course Tuition Subtotal (3 Credits x \$5,500)				16,500.00
Semester Student Activity & Lab Resource Fee:				1,500.00
Grand Total Payable Fee:				18,000.00
Verified Paid Amount (Admin Approved):				- 18,000.00
Remaining Due Balance:				0.00

Submit Semester Fee Payment

Payment Amount (BOT) (*)

e.g. 18500.00

Calculated Amount: \$ 0.00

Payment Method (*)

bkash Online Merchant (0171-XXXXXX)

Transaction ID (TrnID / Bank Scroll No) (*)

e.g. 9148KLM7Q or TKN-2026-EWU-1921

The unique receipt or transaction reference provided by your bank or mobile wallet.

Sender Phone No / Bank Branch

e.g. +88017- or Atfahagar Branch

Student Notes / Remarks (Optional)

e.g. Summer 2026 2nd Installment / Full Fee

Verification Flow:

Upon submission, this payment will be dispatched to the Admin & Accounts Verification Portal with status Pending. Once verified by accounts, your clearance and ledger will be updated immediately.

Submit Payment for Verification

Figure 8 : Payments & Financial Ledger.

FACULTY MENU

- Dashboard**
- My Profile
- Assigned Sections
- Gradebook Management
- Advisee Approval

Dr. Ahmed Hasan

Faculty • 165268910

Faculty Dashboard

Welcome, Dr. Ahmed Hasan

Professor & Chairperson | Department of Computer Science & Engineering • (Head of Department)

17

Assigned Sections

0

Assigned Advisee Students

0

Pre-Advising Submissions

ABI-602

Faculty Office Room

Assigned Course Sections & Capacity

Enter Marks & Grades

COURSE	SEC NO	TIME SLOT	ROOM	ENROLLED / CAP	ACTION
CSE302 Database Systems	Sec 1	Sun-Tue 08:30 AM - 10:00 AM	ABI-502	0 / 35	Manage Grades
CSE303 Database Systems Lab	Sec 1	Sun-Tue 10:10 AM - 11:40 AM	ABI-Lab3	0 / 30	Manage Grades
CSE401 Software Engineering & Design	Sec 1	Sun-Tue 01:30 PM - 03:00 PM	ABI-401	0 / 35	Manage Grades
CSE101 Discrete Mathematics	Sec 1	W 10:10 AM - 11:40 AM & M 10:10 AM - 11:40 AM	429 / 530 (C. Lab-2)	0 / 30	Manage Grades

Quick Management

Open Section Gradebook

Approve Advisee Registrations

Figure 9 : Faculty Dashboard.

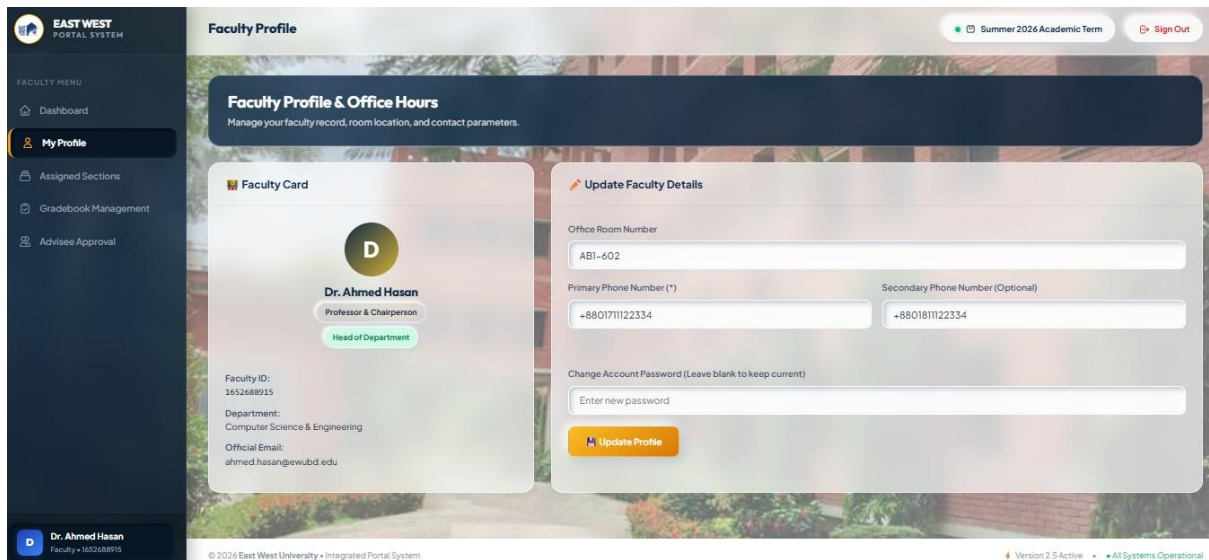


Figure 10 : Faculty Profile.

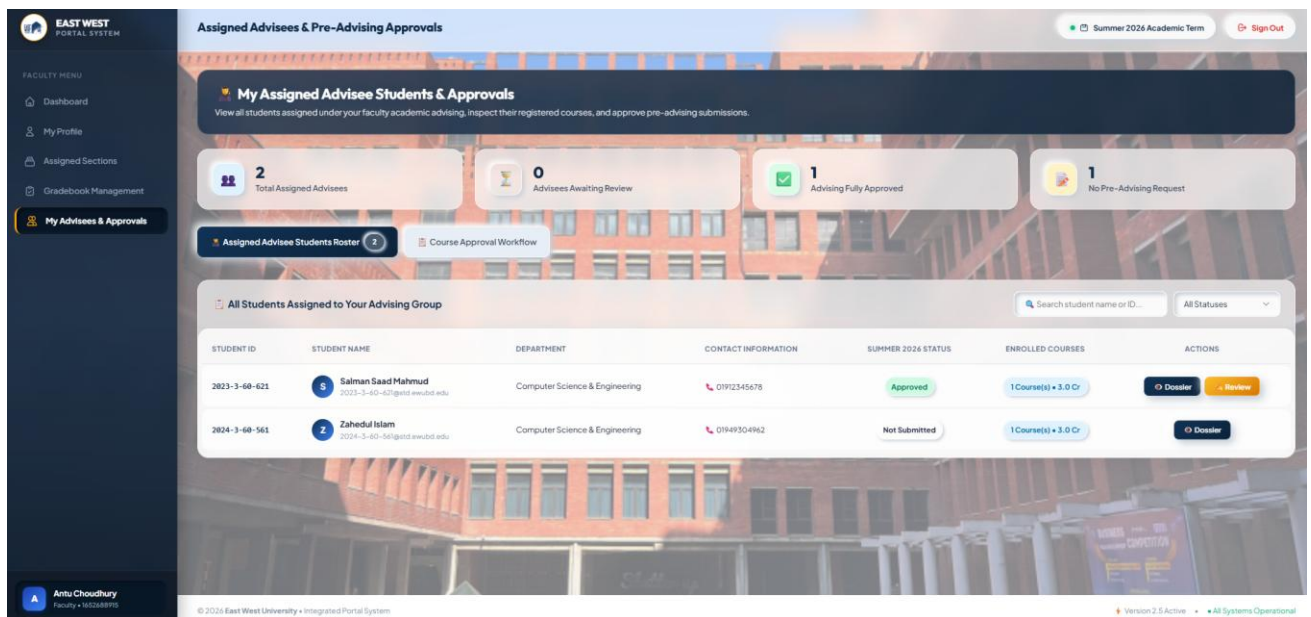


Figure 11 : Advisor Can See Student Details That Will be Advised.

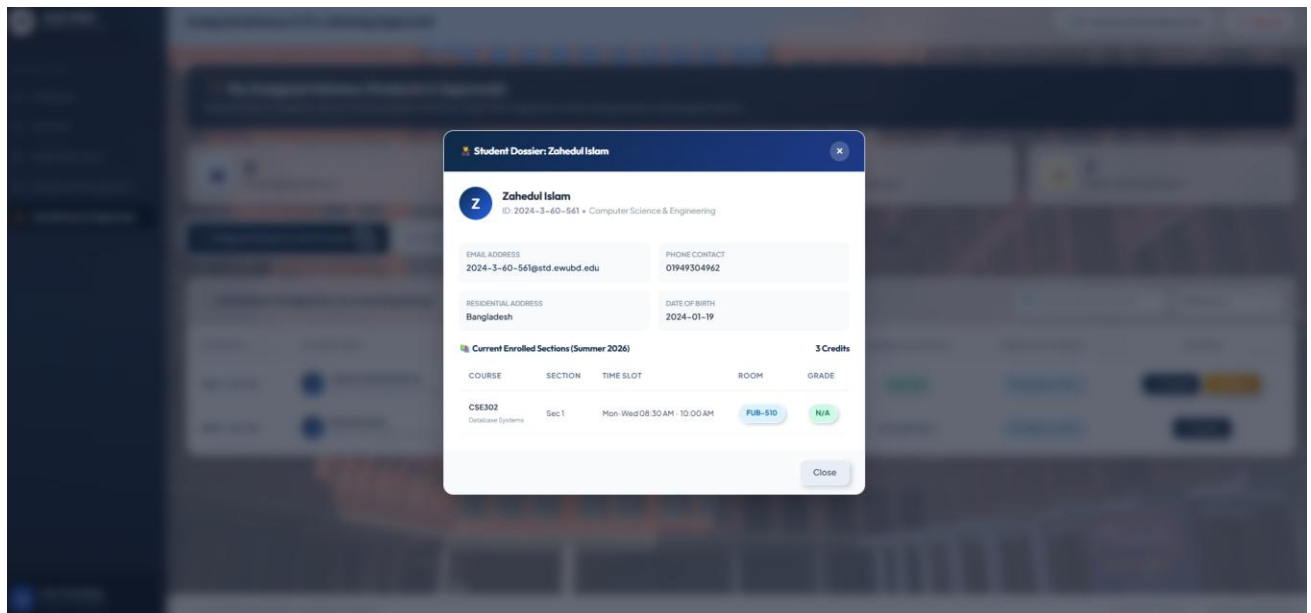


Figure 12 : Advisor Can See Student Details That Will be Advised.

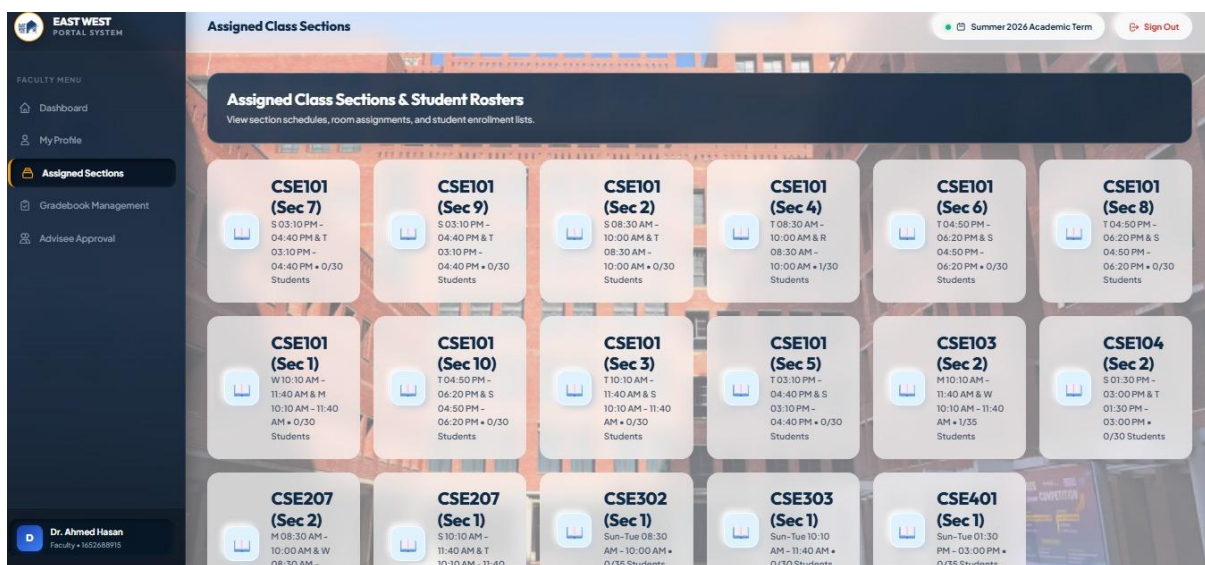


Figure 13 : Assigned Class Sections.

EAST WEST
PORTAL SYSTEM

Gradebook & Evaluation
Summer 2026 Academic Term
Sign Out

Section Gradebook & Mark Evaluation
Enter midterm and final marks, auto-compute official East West University grades (4.00 Scale), and manage advising status.

View EWU Grading Policy

Select Section: CSE302 - Section 1 (Database Systems)

East West University Official Grading Policy

Authorized Scale

NUMERICAL SCORES	LETTER GRADE	GRADE POINT	EVALUATION
80% and above	A+	4.00	Outstanding
75% to less than 80%	A	3.75	Excellent
70% to less than 75%	A-	3.50	Very Good
65% to less than 70%	B+	3.25	Good
60% to less than 65%	B	3.00	Satisfactory

Dr. Ahmed Hasan

Faculty - 1603268915

Figure 14 : Gradebook & Evaluation.

EAST WEST
PORTAL SYSTEM

Advisee Pre-Advising Approval
Summer 2026 Academic Term
Sign Out

Advisee Pre-Advising Approval
Review each student's course requests and approve or reject them one by one.

Assigned Advisees

Courses Awaiting Your Review

Total Submissions

No advisees assigned yet.

The admin office will assign students to you.

Dr. Ahmed Hasan

Faculty - 1603268915

© 2026 East West University • Integrated Portal System

Version 2.5 Active • All Systems Operational

Figure 15 : Advisee Pre-Advising Approved .

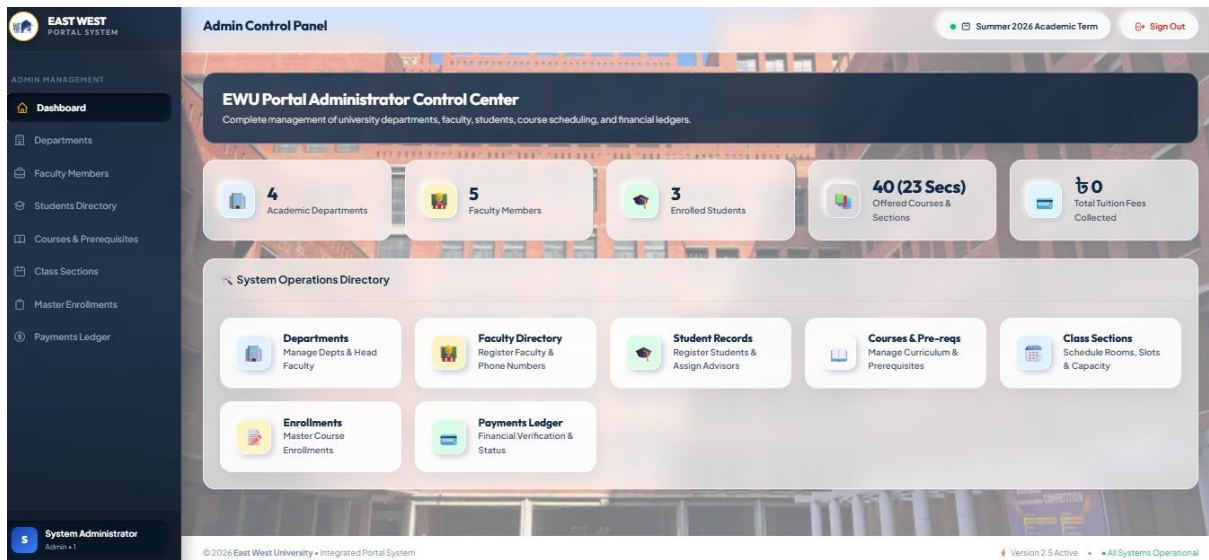


Figure 16 : Admin Control Panel .

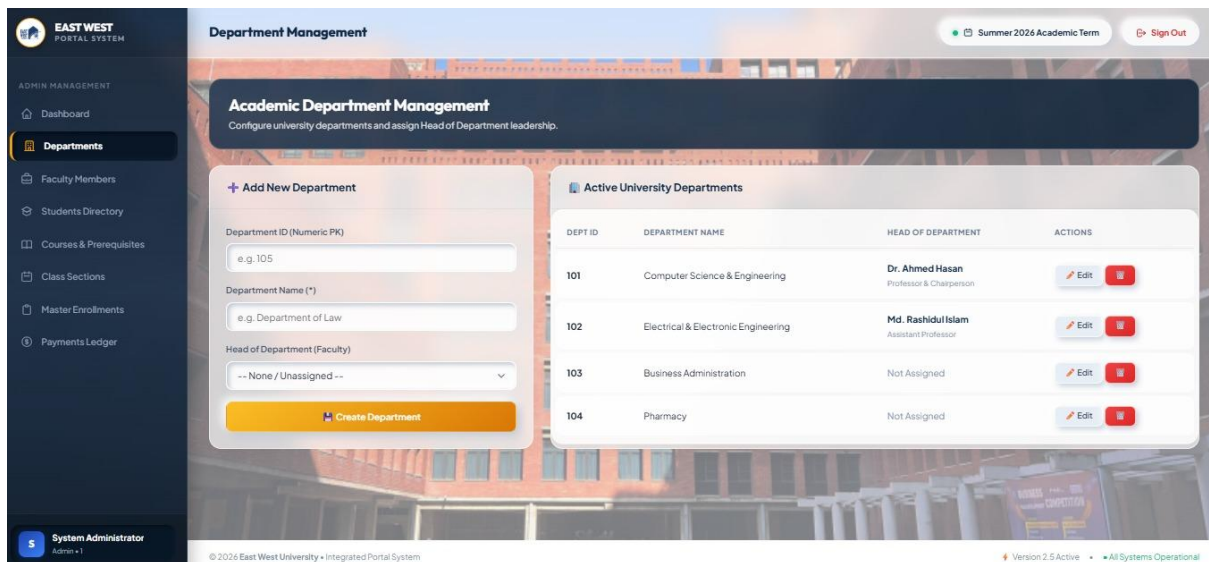


Figure 17 : Department Management .

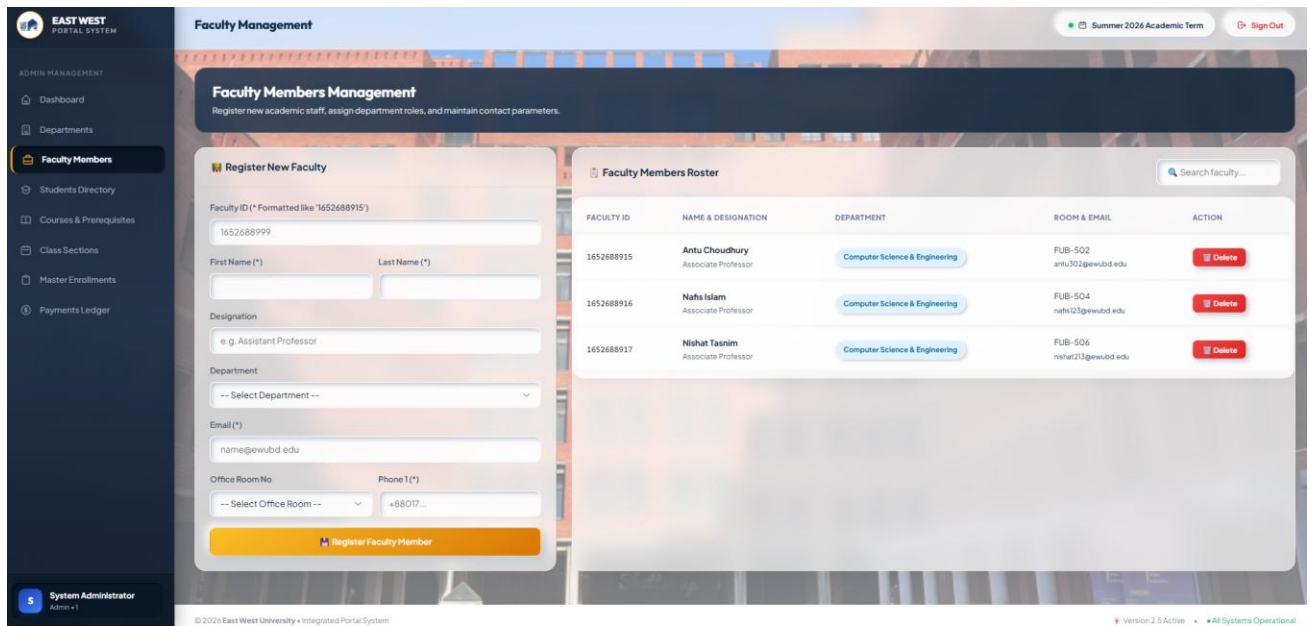


Figure 18 : Faculty Management .

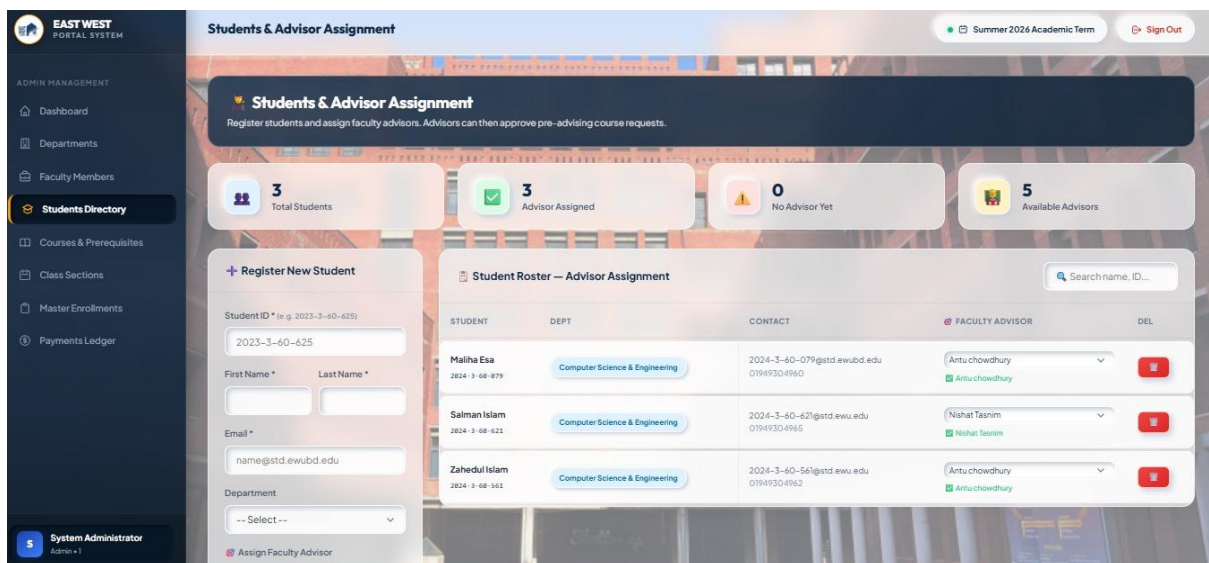


Figure 19 : Students & Advisor Assignment .

ADMIN MANAGEMENT

Dashboard

Departments

Faculty Members

Students Directory

Courses & Prerequisites

Class Sections

Master Enrollments

Payments Ledger

System Administrator

Admin v1

Course & Prerequisite Management

Summer 2026 Academic Term

Sign Out

Academic Courses & Prerequisite Architecture

Manage curriculum subjects, credit weights, and prerequisite dependencies.

Add New Course

Course ID (* e.g. 'CSE-302')

CSE-302

Course Title (*)

Database Systems

Credits (*)

3.0

Department

-- Dept --

Save Course

Link Course Prerequisite

Main Course

-- Select Target Course --

Curriculum Course List

Search course...

COURSE ID	COURSE TITLE	CREDITS	PREREQUISITE REQUIREMENTS	ACTION
BBA101	Principles of Management	3.0	None	
CSE101	Discrete Mathematics	3.0	None	
CSE102	Structured Programming Language	3.0	None	
CSE103	Structured Programming Language	4.0	None	
CSE104	Discrete Mathematics & Logic	3.0	CSE103	
CSE105	Data Structures	3.0	None	
CSE107	Object Oriented Programming	3.0	None	

Figure 20 : Course & Prerequisites Management.

ADMIN MANAGEMENT

Dashboard

Departments

Faculty Members

Students Directory

Courses & Prerequisites

Class Sections

Master Enrollments

Payments Ledger

System Administrator

Admin v1

Section Scheduling & Room Management

Summer 2026 Academic Term

Sign Out

Class Sections & Room Scheduling

Manage course section numbers, assigned classrooms/labs, and timetable slots with automated conflict & duplication prevention.

25 Active Sections

20 Classrooms / Labs Assigned

2 / 880 Total Enrolled Seats

Active Clash & Duplication Shield

Schedule New Section

Course (*)

-- Select Course --

Section No (*)

1

Seat Capacity (*)

35

Room / Lab Number (*)

-- Select Classroom / Lab --

Schedule Time Slot (*)

-- Select Time Slot --

Faculty Instructor

-- Unassigned (TBA) --

Schedule Section

Active Class Sections Directory

Search any section...

Table Filters: All Courses All Rooms All Time Slots idReset

ID	COURSE	SECTION	TIME SLOT	ROOM NO	INSTRUCTOR	CAPACITY	ACTIONS
#19	CSE102 Structured Programming Language	Sec 1	S 08:30 AM - 10:00 AM & T 08:30 AM - 10:00 AM	327 / 327	Antu Choudhury antu.302@ewu.edu	0 / 35	
#20	CSE102 Structured Programming Language	Sec 2	S 10:10 AM - 11:40 AM & T 10:10 AM - 11:40 AM	330 / 330	Antu Choudhury antu.302@ewu.edu	1 / 35	
#21	CSE102 Structured Programming Language	Sec 3	T 11:50 AM - 01:20 PM & R 11:50 AM - 01:20 PM	330 / 330	Antu Choudhury antu.302@ewu.edu	0 / 35	
#22	CSE102 Structured Programming Language	Sec 4	T 01:30 PM - 03:00 PM & R 01:30 PM - 03:00 PM	330 / 330	Antu Choudhury antu.302@ewu.edu	0 / 35	
#23	CSE102 Structured Programming Language	Sec 5	S 01:30 PM - 03:00 PM & T 01:30 PM - 03:00 PM	330 / 517	Antu Choudhury antu.302@ewu.edu	0 / 35	
#24	CSE102 Structured Programming Language	Sec 6	S 03:10 PM - 04:40 PM & T 03:10 PM - 04:40 PM	325 / 528	Antu Choudhury antu.302@ewu.edu	0 / 35	

Figure 21 : Section Scheduling Management .

17 | Page

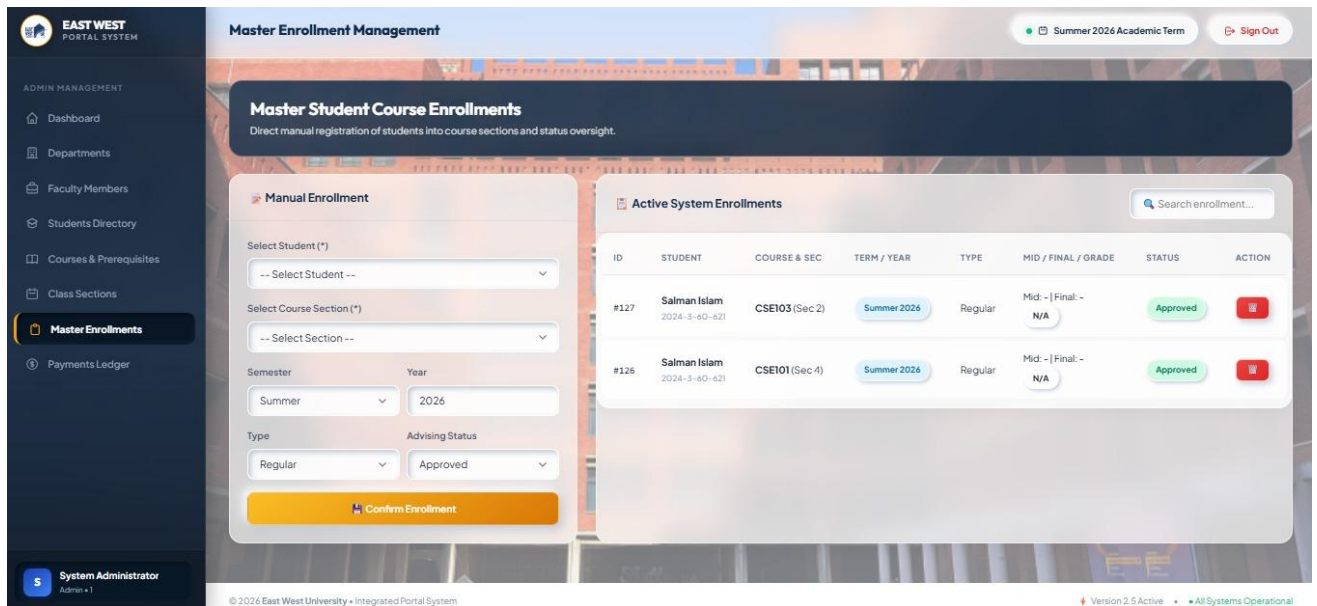


Figure 22 : Master Enrollment Management.

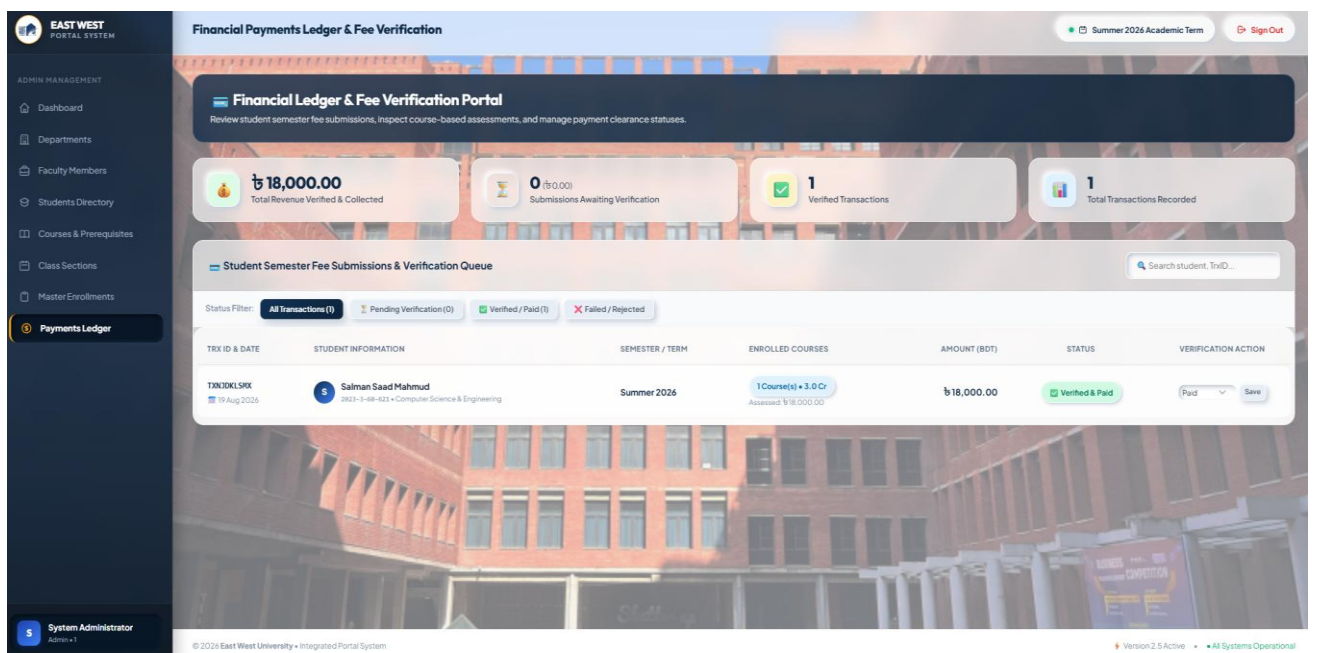


Figure 23 : Financial Payment Ledger .

7. Conclusion

Creating such a portal management system involves creating a fine balance between database normalization and the user experience from an academic perspective. Through optimization of the ER model and simplification of complex processes like advising and prerequisites, the architecture ensures that data integrity is maintained without any need for redundancy. With the addition of role-based security and semester-wise results management features, the system becomes an integrated ecosystem for students, professors, and administrators. In the end, this creates a system that does not create hassles during critical times like advising and grade submissions.

References:

EWU Student Panel: Login <https://share.google/AC1VCNKz6LWTRcz4G>.